

Predicting Breast Cancer Diagnosis Using Machine Learning

ISA 640: Capstone Project

Summer 2026

Milestone 2 Overview

Checkpoint 1 established the project scope, the data pipeline, and the hypothesis that ensemble-based and margin-based classifiers would outperform simpler linear models on this dataset. Milestone 2 builds directly on that foundation: the raw 30-feature dataset is cleaned and profiled, engineered and redundancy-reduced into a final modeling feature set, explored visually to validate the project's hypothesis, and used to train and evaluate five baseline classifiers plus four ensemble strategies.

Handling Missing Values

The dataset was profiled column-by-column for missingness. Consistent with the Checkpoint 1 description of this dataset as curated and complete, all 30 feature columns show 0% missing values, no imputation, deletion, or flagging was required.

Data Imbalance

The target class split is 357 benign (62.7%) to 212 malignant (37.3%) cases; an imbalance ratio of roughly 1.68:1. This is a mild imbalance, not severe enough to warrant synthetic oversampling (e.g., SMOTE) or class-weight rebalancing. Instead, a stratified train/test split (80/20) was used to preserve the class ratio in both partitions, and precision on the malignant class was tracked alongside accuracy throughout model evaluation, since accuracy alone can understate performance on the minority (malignant) class even under mild imbalance.

Features Engineering

Engineered Features

Four new features were engineered from the raw measurements, each motivated by the biological hypothesis that malignant tumors tend to be larger, more irregular in shape, and more heterogeneous than benign ones:

- `worst_mean_radius_ratio`: the ratio of a tumor's "worst" (most extreme) radius to its mean radius, used as a proxy for growth irregularity.
- `concavity_compactness_ratio`: mean concavity relative to mean compactness, isolating boundary "roughness" from overall shape density.
- `radius_concavepoints_interaction`: the product of mean radius and mean concave points, combining size and shape-irregularity into one joint signal.
- `worst_mean_texture_ratio`: the ratio of "worst" to mean texture, capturing texture heterogeneity beyond a simple average.

All four engineered features correlate meaningfully with the malignancy target: `radius_concavepoints_interaction` ($r = 0.733$), `concavity_compactness_ratio` ($r = 0.693$), and `worst_mean_radius_ratio` ($r = 0.641$) show strong correlation, while `worst_mean_texture_ratio` is weaker ($r = 0.173$). Because these correlations are generally weaker than the raw size/shape features themselves, the engineered features were treated as supplementary signal rather than replacements for the original measurements.

Redundant Feature Removal

A pairwise correlation screen (threshold > 0.95) identified 7 redundant base features: area error, mean area, mean perimeter, mean radius, perimeter error, worst area, and worst radius, each dropped in favor of the more target-correlated feature within its correlated group. This reduced the 30 base features to 23, which combined with the 4 engineered features produced a final modeling set of 27 features.

Scaling & Dimensionality Analysis

StandardScaler was applied within model pipelines for scale-sensitive algorithms (Logistic Regression, KNN, SVM), while tree-based models used unscaled inputs. A full Principal Component Analysis (PCA) on the standardized 30-feature set showed that 10 components are sufficient to explain 95% of total variance, confirming substantial redundancy among the original measurements and supporting the feature-reduction approach taken above. A permutation-importance ranking on a Random Forest trained on the final 27-feature set placed worst perimeter as the single strongest predictor, with the engineered worst_mean_radius_ratio and concavity_compactness_ratio also appearing in the top six most important features, evidence that the engineered features are contributing real, if secondary, predictive signal.

Impact of Preprocessing

Since the dataset had no missing values and the class ratio was unchanged by design (a stratified split preserves it), the meaningful comparison is the feature set itself: 30 raw features versus 27 reduced-and-engineered features. To validate that this change was not merely cosmetic, a Logistic Regression baseline was cross-validated (5-fold) on both feature sets:

Stage	Before (Checkpoint 1)	After (Milestone 2)
Missing values	0 (0.0%)	0 (0.0%)
Class ratio (benign:malignant)	1.68 : 1 (357:212)	1.68 : 1 (preserved via stratified split)
Feature count	30	27 (23 base + 4 engineered)
Redundant (>0.95 corr.) pairs	7 present	0 remaining among originals
Scaling	None	StandardScaler (KNN / SVM / LR pipelines)

- 5-fold CV accuracy: 30 raw features: 97.14% ($\pm 1.12\%$)
- 5-fold CV accuracy: 27 reduced + engineered features: 97.58% ($\pm 2.24\%$)

The reduced, de-duplicated feature set matched, and marginally improved on, the full 30-feature baseline, while using fewer and less redundant inputs. This is particularly relevant for linear and distance-based models, where multicollinearity among the dropped radius/perimeter/area features could otherwise inflate coefficient variance without adding predictive value.

EDA & Summary Statistics

Descriptive statistics (count, mean, standard deviation, quartiles) were generated for all 30 original features. Outlier detection used the IQR method, flagging values outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ chosen over a z-score approach because several features (e.g., area, concavity) are right-skewed rather than normally distributed, and z-scores assume approximate normality. The highest outlier rates were concentrated in the “area error” (11.4%), “radius error” and “perimeter error” (6.7% each), and “worst area” (6.2%) features. These are treated as informative extreme values reflecting genuinely more heterogeneous tumors rather than data-entry errors, and were not removed.

Model Training & Evaluation

Using the final 27-feature set, an 80/20 stratified train/test split (455 training / 114 test records) was used to train and evaluate five baseline classifiers, Logistic Regression, K-Nearest Neighbors (k=7), Support Vector Machine (RBF kernel), Decision Tree (max depth 5), and Random Forest (300 trees), followed by four ensemble strategies built on these same base learners: hard/soft Voting, Bagging, AdaBoost, and Gradient Boosting, plus a Stacking classifier with a Logistic Regression meta-learner. Every model was evaluated on test-set accuracy and precision on the malignant class (the diagnostically higher-stakes error to avoid is a false negative, declaring a malignant tumor benign).

Model	Test Accuracy	Precision (Malignant)
Ensemble - Gradient Boosting	0.9825	1.0000
Ensemble - Stacking	0.9825	1.0000
Ensemble - Voting (Soft)	0.9825	1.0000
Ensemble - AdaBoost	0.9825	1.0000
Random Forest (n=300)	0.9737	1.0000
Ensemble - Bagging (DT base)	0.9737	1.0000
SVM (RBF)	0.9649	0.9750
Ensemble - Voting (Hard)	0.9649	1.0000
KNN (k=7)	0.9561	1.0000
Logistic Regression	0.9474	0.9737
Decision Tree (max_depth=5)	0.9035	0.9429

Random Forest (97.37% accuracy, 100% malignant precision) was the strongest single (non-ensemble) model, consistent with the hypothesis that ensemble methods would outperform simpler linear approaches, Logistic Regression trailed at 94.74% accuracy. The Decision Tree was the weakest performer overall (90.35% accuracy), reflecting the known tendency of a single shallow tree to underfit relative to an ensemble of trees. SVM (RBF) reached 96.49% accuracy, in line with the 96.8% figure cited from prior literature in Checkpoint 1. The ensemble methods pushed performance further still: Gradient Boosting, Stacking, Soft Voting, and AdaBoost tied for the best overall result at 98.25% accuracy with 100% precision on the malignant class (only 2 of 42 malignant test cases misclassified as benign, and zero benign cases misclassified as malignant).

Key Findings

- Ensemble methods outperformed simpler linear and single-tree models, confirming the project's hypothesis: the four best-performing models (Gradient Boosting, Stacking, Soft Voting, AdaBoost) were all ensemble approaches, each reaching 98.25% test accuracy.
- Random Forest was the strongest standalone (non-ensemble) model at 97.37% accuracy with perfect precision on the malignant class, reinforcing that this dataset's non-linear feature relationships favor tree-based ensembles.
- Size and shape-irregularity measurements (worst perimeter, worst concave points, worst radius, worst area) consistently emerged as the strongest predictors of malignancy across correlation analysis, permutation importance, and PCA loadings, directly answering the project's first supporting question.
- Feature engineering and redundancy reduction (30 to 27 features) preserved, and marginally improved, baseline model accuracy while reducing multicollinearity, validating the preprocessing approach taken in Section 3.
- Precision on the malignant class remained at or near 100% for every ensemble model and for Random Forest, indicating very few false negatives (missed malignant diagnoses), the error type with the highest clinical cost.